

1 Solution — GIAM 9.1

Problem 2

1.1 Problem

In this section we talked about similarity. Two figures in the plane are similar if it is possible to turn one into the other by a sequence of mappings: a translation, a rotation, and a scaling.

Geometric similarity is an equivalence relation. To fix our notation, let $T(x, y)$ represent a generic translation, $R(x, y)$ a rotation, and $S(x, y)$ a scaling — thus a generic similarity is a function from $\mathbb{R}^2$ to $\mathbb{R}^2$ that can be written in the form $S(R(T(x, y)))$.

Discuss the three properties of an equivalence relation (reflexivity, symmetry, and transitivity) in terms of geometric similarity.

1.2 Setup — What a Similarity Really Looks Like

Pin down the elementary pieces:

- A *translation* T_v sends $p \mapsto p + v$, where $v \in \mathbb{R}^2$.
- A *rotation* R_θ about the origin sends $p \mapsto R_\theta p$, where R_θ is the linear map “rotate by angle θ ”.
- A *scaling* S_k about the origin sends $p \mapsto kp$, where $k > 0$ is the scale factor.

A generic similarity has the form $\varphi = S_k \circ R_\theta \circ T_v$, i.e.

$$\varphi(p) = k \cdot R_\theta(p + v) = kR_\theta p + kR_\theta v.$$

Set $w := kR_\theta v$ — the “effective output translation”. Then

$$\varphi(p) = kR_\theta p + w.$$

As v varies over $\mathbb{R}^2$, so does w (since $k > 0$ and R_θ is invertible). So *every* map of this form is a similarity, and conversely.

Working form. A similarity is a map $\varphi : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ of the form

$$\varphi(p) = kR_\theta p + w, \quad k > 0, \theta \in \mathbb{R}, w \in \mathbb{R}^2.$$

Two figures $F_1, F_2 \subseteq \mathbb{R}^2$ are **similar** (written $F_1 \sim F_2$) if there is a similarity φ with $\varphi(F_1) = F_2$.

1.3 Reflexivity — $F \sim F$

We need a similarity that fixes every figure F . The identity map $p \mapsto p$ does it — and it is a similarity: take $k = 1$, $\theta = 0$, $w = 0$:

$$\varphi(p) = 1 \cdot R_0 \cdot p + 0 = p.$$

In the original $S \circ R \circ T$ form: $T = T_0$ (translate by zero), $R = R_0$ (rotate by zero), $S = S_1$ (scale by one). Composition is the identity.

So $F \sim F$ for every figure F . ■

1.4 Symmetry — If $F_1 \sim F_2$ Then $F_2 \sim F_1$

Suppose $F_1 \sim F_2$ via $\varphi(p) = kR_\theta p + w$. We want a similarity ψ with $\psi(F_2) = F_1$. The natural candidate is $\psi = \varphi^{-1}$.

Solving $q = kR_\theta p + w$ for p :

$$\begin{aligned} \varphi^{-1}(q) &= R_\theta^{-1} \left(\frac{q-w}{k} \right) \\ &= \frac{1}{k} R_{-\theta} (q - w) \\ &= \frac{1}{k} R_{-\theta} q - \frac{1}{k} R_{-\theta} w. \end{aligned}$$

This has the working form $k'R_{\theta'}q + w'$ with

$$k' = \frac{1}{k} > 0, \quad \theta' = -\theta, \quad w' = -\frac{1}{k} R_{-\theta} w \in \mathbb{R}^2.$$

So φ^{-1} is a similarity, and $\varphi^{-1}(F_2) = F_1$. Thus $F_2 \sim F_1$. ■

In the original $S \circ R \circ T$ form: φ^{-1} decomposes as $S_{1/k} \circ R_{-\theta} \circ T_{-w}$:

$$(S_{1/k} \circ R_{-\theta} \circ T_{-w})(q) = \frac{1}{k} R_{-\theta} (q - w) = \varphi^{-1}(q). \quad \checkmark$$

1.5 Transitivity — If $F_1 \sim F_2$ and $F_2 \sim F_3$ Then $F_1 \sim F_3$

Suppose $\varphi_1(p) = k_1 R_{\theta_1} p + w_1$ maps F_1 to F_2 , and $\varphi_2(p) = k_2 R_{\theta_2} p + w_2$ maps F_2 to F_3 . Their composition maps F_1 to F_3 .

Compute:

$$\begin{aligned} (\varphi_2 \circ \varphi_1)(p) &= \varphi_2(k_1 R_{\theta_1} p + w_1) \\ &= k_2 R_{\theta_2} (k_1 R_{\theta_1} p + w_1) + w_2 \\ &= (k_1 k_2) R_{\theta_2} R_{\theta_1} p + k_2 R_{\theta_2} w_1 + w_2 \\ &= (k_1 k_2) R_{\theta_1 + \theta_2} p + (k_2 R_{\theta_2} w_1 + w_2). \end{aligned}$$

This has the working form $k' R_{\theta'} p + w'$ with

$$k' = k_1 k_2 > 0, \quad \theta' = \theta_1 + \theta_2, \quad w' = k_2 R_{\theta_2} w_1 + w_2 \in \mathbb{R}^2.$$

So $\varphi_2 \circ \varphi_1$ is a similarity, and $F_1 \sim F_3$. ■

Key fact used. $R_{\theta_2} R_{\theta_1} = R_{\theta_1 + \theta_2}$ — composing two rotations about the same center gives a rotation whose angle is the *sum* of the two angles.

1.6 Conclusion — Similarities Form a Group

The three equivalence-relation properties of $\sim$ are exactly the three group-theoretic facts about similarity maps:

Equivalence property of $\sim$	Underlying fact about similarity maps
Reflexivity	The identity is a similarity.
Symmetry	The inverse of a similarity is a similarity.
Transitivity	The composition of two similarities is a similarity.

Table 1.1

Together these say exactly that **similarities of the plane form a group** under composition. *Equivalence under a group action is always an equivalence relation* — that's the

structural reason $\sim$ behaves as it does, and the calculations above are merely the verification.

1.7 Remark — Why the $S \circ R \circ T$ Form is Enough

The textbook's $S(R(T(x, y)))$ form might seem restrictive: it pins down a specific order (translate, then rotate, then scale). Symmetry and transitivity, however, produce expressions like $T_{-w} \circ R_{-\theta} \circ S_{1/k}$ or $S \circ R \circ T \circ S \circ R \circ T$ — orderings that don't *look* canonical.

The redemption: rotations, scalings, and translations don't all commute, but they *almost* commute — composition can always be rewritten in the canonical order at the cost of adjusting parameters. The relevant identities:

$$\begin{aligned} S_k \circ T_v &= T_{kv} \circ S_k, \\ R_\theta \circ T_v &= T_{R_\theta v} \circ R_\theta, \\ S_k \circ R_\theta &= R_\theta \circ S_k. \end{aligned}$$

The first says scaling-then-translating is the same as translating-by-a-scaled-vector first, then scaling. The second is the analogue for rotation. The third — scalings and rotations *about the origin* — genuinely commute.

These three identities let us shuffle a long composition into the canonical $S_k \circ R_\theta \circ T_v$ shape one step at a time: push all translations to the right, push all scalings to the left, and you're done.

1.8 Remark — Direct vs Indirect Similarities

The textbook's definition silently excludes **reflections**. A similarity that includes a reflection (e.g. $(x, y) \mapsto (x, -y)$) reverses *orientation*. Maps of the form $S \circ R \circ T$ — the textbook's form — are all **direct (orientation-preserving) similarities**.

If we admitted reflections, $\sim$ would still be an equivalence relation — the three arguments above go through with one extra check (a reflection is its own inverse, $\mathbf{Refl} \circ$

$\text{Refl} = \text{id}$, so the symmetry property is even simpler). Two figures that are *mirror images* would then be identified — a useful convention in elementary geometry.

For Conway’s Morley proof in Section 9.1, only direct similarities matter: the entire construction sits in a single oriented plane, and the similarity from the assembled puzzle back to $\triangle ABC$ is built out of translation, rotation, and scaling alone — no flip is needed.

1.9 Remark — Algebraic Shortcut via $\mathbb{C}$

Identify $\mathbb{R}^2$ with $\mathbb{C}$ via $(x, y) \leftrightarrow x + iy$. Every direct similarity has the *single-variable complex* form

$$\varphi(z) = cz + w, \quad c, w \in \mathbb{C}, c \neq 0.$$

The complex number $c = k e^{i\theta}$ packages the scale factor and rotation angle into one quantity (its modulus is k , its argument is θ).

In this language the three checks collapse to one-line arithmetic:

Property	Witness in $\mathbb{C}$
Reflexivity	$\varphi(z) = z$ with $c = 1$, $w = 0$.
Symmetry	$\varphi^{-1}(z) = z/c - w/c$, so $c' = 1/c$ and $w' = -w/c$.
Transitivity	$(\varphi_2 \circ \varphi_1)(z) = c_2 c_1 z + (c_2 w_1 + w_2)$.

Table 1.2

Each row is literally the same calculation done above over $\mathbb{R}^2$, with rotation-by- θ replaced by multiplication by $e^{i\theta}$.

1.10 Remark — Similarities are Bijections That Scale All Distances Equally

Each similarity $\varphi(p) = kR_\theta p + w$ enjoys three properties that make the term “similar” honest:

- **Bijjective**, with inverse $\varphi^{-1}(q) = \frac{1}{k}R_{-\theta}(q - w)$.
- **Continuous** (in fact Lipschitz with constant k).
- **Distance-scaling by exactly k** : $\|\varphi(p) - \varphi(q)\| = k \|p - q\|$ for all p, q .

It is the third property that justifies the *name*: φ multiplies every length by the same factor k , so shapes are preserved — angles unchanged, ratios of side lengths unchanged.

When $k = 1$, a similarity is an **isometry** (a rigid motion), and the induced equivalence relation is *congruence* — a finer equivalence than similarity, since congruent figures must additionally be the same *size*.

1.11 Remark — What “Discuss” Buys You Pedagogically

The three properties of an equivalence relation feel abstract when stated as bare axioms (reflexive, symmetric, transitive). Translating them into geometric similarity gives each property a *picture*:

- **Reflexivity** is the trivial observation that “ F is similar to itself” — the identity transformation is always available.
- **Symmetry** is “I can undo any similarity by another similarity” — every shape-preserving transformation has a shape-preserving inverse.
- **Transitivity** is “chaining shape-preserving transformations gives a shape-preserving transformation” — and shape (angles and length-ratios) is preserved under composition.

These three together are what justify lumping all triangles with the same three angles into a single *similarity class* — and ultimately what lets Conway “build a triangle similar to $\triangle ABC$ ” in Section 9.1 and conclude that the *original* triangle inherits the equilateral-Morley property.